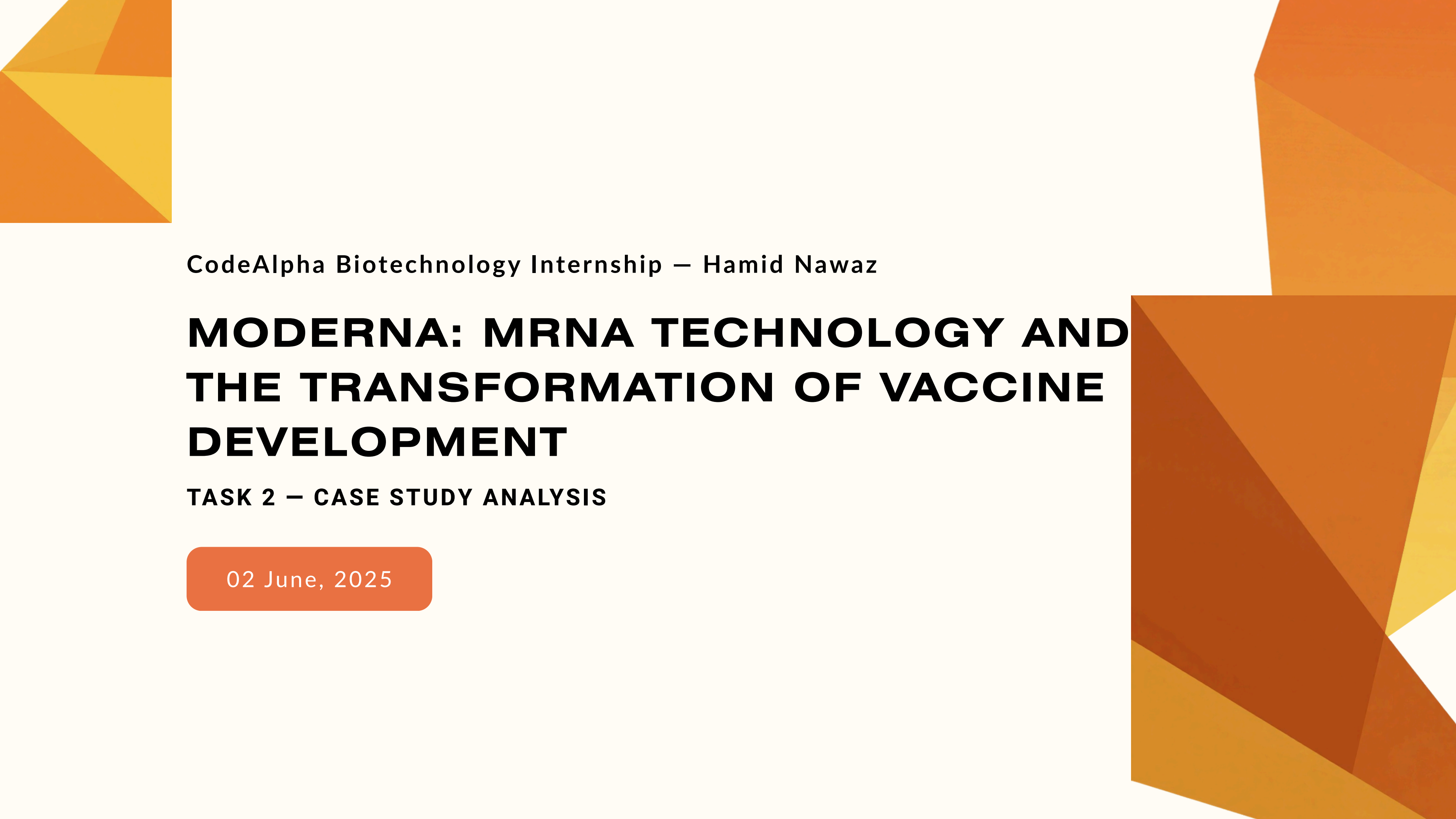




CodeAlpha Biotechnology Internship – Hamid Nawaz

MODERNA: MRNA TECHNOLOGY AND THE TRANSFORMATION OF VACCINE DEVELOPMENT

TASK 2 – CASE STUDY ANALYSIS

02 June, 2025

COMPANY INTRODUCTION

Founded in 2010 in Cambridge, Massachusetts, Moderna began as a bold idea: harness messenger RNA to instruct the body's own cells to produce therapeutic proteins.

Timeline & Evolution: 2010 — Founded by Robert Langer, Derrick Rossi, and others; 2013–2018 — Early mRNA research, strategic partnerships with DARPA and BARDA; 2018 — IPO, becoming the largest biotech IPO at the time; 2020–2021 — Emergency Use Authorization for mRNA-1273 (Spikevax) COVID-19 vaccine; 2022–2025 — Expansion into RSV, influenza, oncology, and rare diseases.

Mission: To deliver on the promise of mRNA science to create a new generation of medicines. Moderna operates as a platform company — its mRNA technology is not a single product but a programmable foundation enabling rapid development across multiple disease areas, from infectious diseases to personalized cancer vaccines.



THE BIOTECHNOLOGY PROBLEM

LIMITATIONS OF CONVENTIONAL VACCINE DEVELOPMENT

Traditional vaccine development relies on growing live or inactivated pathogens, a process that is slow, costly, and highly pathogen-specific. Vaccine production timelines typically span 10–15 years from discovery to approval. Manufacturing requires specialized biosafety facilities, and scaling up for global distribution is a major bottleneck. Each new pathogen demands an entirely new production process, making rapid response nearly impossible.

THE NEED FOR RAPID RESPONSE TO EMERGING PATHOGENS

Emerging infectious diseases such as SARS, MERS, Ebola, and COVID-19 have exposed critical gaps in the world's ability to respond quickly to novel threats. Conventional platforms cannot be rapidly reprogrammed for new targets. The lag between pathogen identification and vaccine availability costs lives. There was a clear need for a flexible, programmable, and rapidly scalable vaccine platform capable of responding to unknown future pathogens.

MODERNA'S CORE INNOVATIONS

Moderna's mRNA platform integrates four pillars into a unified, programmable technology engine – enabling rapid, scalable, and adaptable vaccine and therapeutic development.

01

MRNA PLATFORM

Synthetic mRNA instructs cells to produce target proteins. One platform, infinite programmable targets – no live pathogens or eggs required in production.

02

LNP DELIVERY SYSTEM

Lipid nanoparticles encapsulate and protect mRNA, enabling cellular uptake. LNP formulations are reusable across products, accelerating new vaccine development.

03

DIGITAL SEQUENCE DESIGN

Antigen sequences are digitally designed and optimized in silico. This enables rapid response to new pathogens and supports scalable, automated manufacturing pipelines.

HOW MRNA TECHNOLOGY WORKS

STEP 1–2

Step 1 – DNA Blueprint: Pathogen gene sequence is identified and digitally mapped.
Step 2 – mRNA Design: Synthetic mRNA strand is engineered in silico encoding the target antigen (e.g., spike protein).

STEP 3–4

Step 3 – LNP Encapsulation: Designed mRNA is encapsulated in lipid nanoparticles (LNPs) to protect it and enable cellular entry.
Step 4 – Cell Uptake: LNPs fuse with cell membranes; mRNA is delivered into the cytoplasm of host cells.

STEP 5–6

Step 5 – Protein Production: Ribosomes read the mRNA and produce the target antigen protein (e.g., prefusion spike).
Step 6 – Immune Response: Immune system detects the foreign antigen, producing antibodies and memory T/B cells.

STEP 7

Step 7 – mRNA Degradation: mRNA is naturally broken down by cellular enzymes within days – no DNA alteration occurs.
Result: Durable immune memory is established without any live pathogen involved.



MAJOR PRODUCT & PROOF OF CONCEPT

SPIKEVAX (MRNA-1273): FDA APPROVAL & VALIDATION

Spikevax (mRNA-1273) is Moderna's COVID-19 vaccine and the platform's definitive proof of concept. It encodes a stabilized prefusion spike antigen using nucleoside-modified mRNA encapsulated in lipid nanoparticles (LNPs). Granted FDA Emergency Use Authorization in December 2020 and full FDA approval (Biologics License Application) in January 2022, Spikevax demonstrated 94.1% efficacy in Phase 3 trials – confirming that mRNA-LNP technology could produce a safe, highly effective vaccine at unprecedented speed.

WHY IT VALIDATED THE PLATFORM

Spikevax proved that mRNA instructions alone – without live virus – could trigger robust immune responses. The LNP delivery system protected fragile mRNA from degradation and enabled efficient cellular uptake. Crucially, when variants emerged, Moderna rapidly updated the mRNA sequence to produce bivalent boosters (e.g., BA.4/BA.5), demonstrating the platform's core advantage: sequence-driven adaptability at scale. This validated mRNA-LNP as a reusable, programmable vaccine engine – not a one-time product.

SCIENTIFIC INNOVATION

PRODUCT EXPANSION

mRESVIA: First mRNA-based RSV vaccine for adults 60+

mNEXSPIKE: Updated COVID-19 mRNA booster (UK/EU)

Platform speed enables rapid sequence redesign

KEY PRODUCTS & 2026 INNOVATION DIRECTION



MRESVIA (RSV)

FDA-Approved mRNA
vaccine for RSV



MNEXSPIKE

Updated mRNA COVID
Booster Platform



COMBINATION VACCINES

Multiplex Flu+COVID
mRNA Pipeline 2026

APPLICATIONS BEYOND COVID-19

01

INFECTIOUS DISEASES

02

IMMUNO-ONCOLOGY

03

PERSONALIZED CANCER VACCINES

04

RARE GENETIC DISEASES

05

AUTOIMMUNE DISEASES

06

APPROVED PRODUCTS

07

INVESTIGATIONAL PIPELINE

08

PLATFORM VERSATILITY

CHALLENGES

01

REACTOGENICITY & SAFETY

02

DELIVERY & TISSUE TARGETING

03

DURABILITY & BOOSTING

04

MANUFACTURING & COLD-CHAIN

05

REGULATORY COMPLEXITY

06

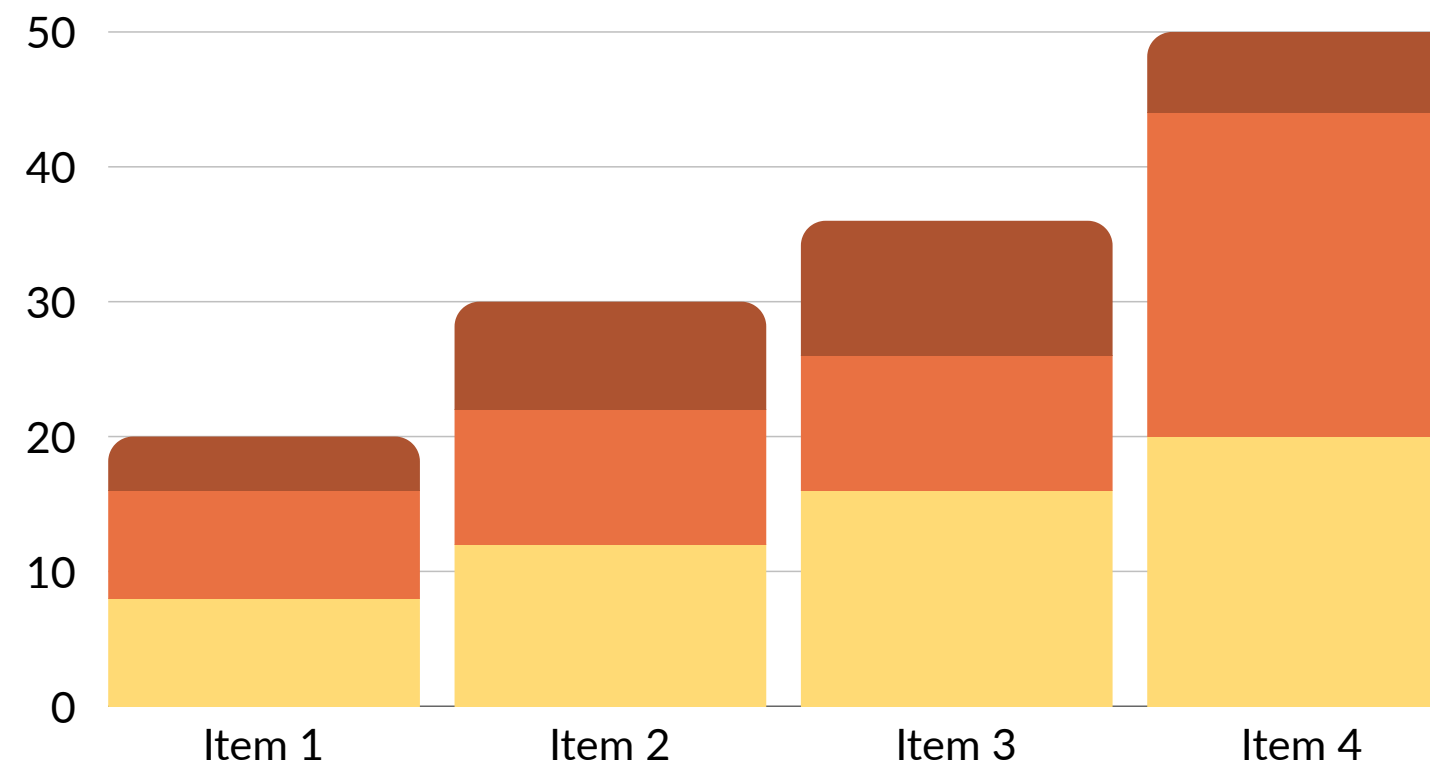
COMPETITION & MARKET PRESSURE

07

POST-PANDEMIC DEMAND

08

EQUITABLE GLOBAL ACCESS



MARKET IMPACT

COVID-19 proved the global scalability of mRNA-LNP vaccines, with Moderna delivering hundreds of millions of doses worldwide. The pandemic catalyzed a new vaccine platform category, attracting billions in investment and reshaping the biotech industry. Moderna transitioned from a pre-revenue startup to a multi-billion-dollar commercial biotech, validating mRNA as a foundational technology for future infectious disease and oncology applications.

FUTURE PROSPECTS (2026 VIEW)

Moderna's expanding pipeline leverages its mRNA platform across respiratory, oncology, and rare disease programs, accelerated by AI and automation.

01

RESPIRATORY PIPELINE

Moderna expects to advance flu, flu+COVID combination, and norovirus vaccines. These represent near-term commercial opportunities leveraging platform speed and adaptability. (Forward-looking statements reflect company expectations.)

02

ONCOLOGY & RARE DISEASE

Investigational programs in personalized cancer vaccines (mRNA-4157/V940) and rare disease enzyme replacement therapies expand Moderna beyond infectious disease into high-value therapeutic modalities.

03

AI & AUTOMATION

Moderna integrates AI, machine learning, robotics, and scientific intelligence to accelerate sequence design, candidate selection, and manufacturing optimization, shortening development timelines significantly.

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CONCLUSION & KEY TAKEAWAYS

Hamid Nawaz